Telecom Churn Profit Optimization

How can we transform customer behavior data into a self-funding retention engine?

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Problem Statement

Revenue Recovery Through Profit-Optimized Retention

01. The Challenge: Revenue Leakage

- **The Cost:** It is **5x more expensive** to acquire a new customer than to retain an existing one.
- **The Drain:** Every churned user represents a **~400 KES loss** in Lifetime Value (LTV).
- **The Gap:** Traditional models focus on "Accuracy," ignoring the **ROI of the intervention budget.**

02. The Insight: "Service Fatigue"

- **The Trigger:** Data reveals a "Boiling Point" at **3+ customer service calls**.
- **The Risk:** High-usage International Plan users are **3x more likely** to exit due to price sensitivity.

03. The Strategy: The Profit "Sweet Spot"

- **The Solution:** A Decision Tree model tuned to a **0.7 probability threshold**.
- **The Impact:** * ✅ **Eliminated 70% of budget waste** on loyal customers.
    - ✅ **Maximized Net Profit** by targeting only "at-risk" high-value users.

Objectives

Shifting from "Model Accuracy" to "Business Value Optimization"

1. Predictive Behavioral Modeling

- **The Intelligence: Developed a Decision Tree Classifier to map the "Churn DNA" of our customer base.**
- **The Trigger: Identified "Service Fatigue"—a critical behavioral boiling point occurring at 3+ customer service calls where churn probability spikes.**

2. Revenue Recovery Strategy

- **The Target: Focused on high-value segments, specifically International Plan users, who represent the highest flight risk due to price sensitivity.**
- **The Mitigation: Shifted from reactive "save" attempts to proactive interventions *before* the customer reaches the exit point.**

3. Business Value Optimization

- **The Sweet Spot: Implemented a 0.7 Probability Threshold for intervention.**
- **The Efficiency: Instead of a "blanket" 0.5 threshold, this selective approach ensures we only spend the 100 KES retention budget on customers who are statistically likely to leave but possible to save.**

4. Financial Impact Evaluation

- **The ROI: Every decision is measured against a Business Value Function.**
- **The Result: Minimized "False Positives" (wasted vouchers) while maximizing "True Positives" (saved revenue), resulting in a self-funding retention campaign.**

Dataset Overview

I used “**Churn in Telecom's dataset”**

Transforming 21 raw variables into actionable "Churn Predictors."

01. Customer Usage Patterns (The "How")

- **The Indicators:** Total day, evening, and night minutes, along with associated charges.
- **The Strategy:** Analyzed the **Volume of Activity** vs. **Cost Sensitivity**.
- **Key Finding:** High daytime usage combined with high charges created a "Price Pressure" segment.

02. Service Interaction Log (The "Why")

- **The Variable: Customer Service Calls.**
- **The Insight:** This is the most powerful "Signal" in the dataset.
- **The Boiling Point:** Beyond **3 interactions**, a customer is no longer just "seeking help"—they are expressing **"Service Fatigue"** and preparing to exit.

03. Feature Segments (The "Who")

- **Plan Dynamics:** Compared **International Plan** users against **Voicemail Plan** users.
- **The Logic:** International Plan users are high-value but highly mobile—they churn at a significantly higher rate if a competitor offers a better rate.

04. The Mission (The Target)

- **The Goal:** Predict **Binary Churn** (0 = Stay | 1 = Leave).
- **The Challenge:** Addressing **Class Imbalance** (only ~15% of customers leave), requiring the use of **SMOTE** to ensure the model could accurately "see" the churners.

Feature Engineering: Decoding the Customer Mindset

Transforming Raw Metrics into Behavioral Predictors

1. Price Sensitivity (`charge_per_min`)

- **The Logic:** Calculated the average cost per minute across all time slots.
- **The Insight:** High-usage customers are often the most **price-sensitive**. By isolating the "unit cost," the model can identify who is most likely to switch to a competitor for a better rate.

2. Customer Frustration (`high_service_user`)

- **The Logic:** Created a binary flag for any customer with **3+ service interactions**.
- **The Insight:** This turned a simple "count" into a **"Friction Signal."** It allows the model to separate a "curious" customer from a "frustrated" one who is actively looking for the exit.

3. Usage Patterns (`day_night_ratio`)

- **The Logic:** A ratio comparing daytime minutes to nighttime/evening usage.
- **The Insight:** Identifies **Professional vs. Personal** users. Professional (Daytime) users have different churn triggers (reliability) compared to Personal (Nighttime) users (cost).

Addressing Class Imbalance: The SMOTE Strategy

Ensuring the model "sees" the churner in a sea of loyalists.

1. The Imbalance Challenge

- **The Data Reality:** Only **~15%** of our dataset represents customers who churned.
- **The "Accuracy" Trap:** A standard model could reach **85% accuracy** by simply predicting that *no one* ever leaves. This would be mathematically "correct" but business-useless.

2. The SMOTE Solution (Synthetic Minority Over-sampling Technique)

- **The Logic:** Instead of just duplicating data, SMOTE creates **synthetic examples** of churners by analyzing the "neighborhood" of existing churn data points.
- **The Result:** It balances the training set (50/50), forcing the algorithm to learn the specific behavioral patterns—like high service calls or price sensitivity—that lead to an exit.

3. Why This Matters for Profit

- **Sensitivity vs. Specificity:** By balancing the data, we significantly improved the model's **Recall** (its ability to catch actual churners).
- **The Bottom Line:** You cannot save a customer the model fails to see. SMOTE ensures our "Revenue Recovery" engine is tuned to detect the quietest signals of frustration.

The Modeling Strategy: From Logic to Decisions

Balancing Interpretability with Predictive Power

1. The Candidates

- **Logistic Regression (The Baseline):** * **The Role:** Used as our "control" model to establish a performance floor.
    - **The Benefit:** High transparency. It allowed us to see the exact "weight" each feature (like service calls) had on the probability of churn.
- **Decision Tree (The Strategic Champion):**
    - **The Role:** Our primary engine for segmenting customers.
    - **The Benefit:** Perfectly handles non-linear relationships. It excels at identifying the "Thresholds" (e.g., exactly at **3.5 calls**, the risk profile shifts).

2. The Evaluation Framework

- **Beyond Accuracy:** Traditional metrics like "Accuracy" are misleading in churn. We focused on **Recall** (to catch as many churners as possible) and **Precision** (to avoid wasting vouchers on loyalists).
- **The Business Value Filter:** We didn't stop at a "Yes/No" prediction. We passed every prediction through a **Cost-Benefit Matrix** to ensure the model was actually generating a profit, not just a label.

3. The Selection Logic

- **Winner:** The **Decision Tree** was selected for final deployment.
- **Why:** It provided the clearest "Rules of Engagement" for the marketing team, allowing them to see exactly which customer segments (High usage + High service calls) were the most profitable to save.

Business Evaluation: The Economics of Retention

Maximizing Net Profit through Targeted Intervention

1. The Financial Parameters

- **Customer Value (LTV):** Each retained customer represents **400 KES** in recovered Lifetime Value.
- **Cost of Action:** Each proactive retention offer (e.g., a service voucher) costs the company **100 KES**.
- **Intervention Success:** Based on historical trends, we assume a **40% success rate**—meaning for every 10 customers we target, 4 are successfully "saved" from churning.

2. The Profit & Loss Matrix

To find the "Sweet Spot," we evaluated every model prediction through three lenses:

- ✅ **True Positive (The Win):** We correctly identified a churner. The **Profit** is the saved 400 KES minus the 100 KES cost.
- ⚠️ **False Positive (The Waste):** We sent a voucher to a loyal customer who wasn't leaving. The **Cost** is a wasted 100 KES.
- ❌ **False Negative (The Leak):** We missed a churner. The **Loss** is the full 400 KES in future revenue.

3. Strategic Deployment

- **The Threshold:** We don't just act on a 50/50 guess.
- **The Optimization:** By shifting our intervention threshold to **0.7**, we ensure that we only spend the 100 KES budget when the probability of a "Save" is high enough to guarantee a **Net Positive ROI**.

Financial Results: Comparing Strategy Performance

Benchmarking Net Impact Across Models

1. The Performance Gap

- **Baseline (Logistic Regression):** A net impact of **-15,880 KES**. This model was too "sensitive," sending vouchers to too many customers who weren't actually going to leave (High False Positives).
- **Champion (Decision Tree):** A net impact of **-9,300 KES**. By better segmenting the "Service Fatigue" and "International Plan" users, this model **reduced the financial loss by 41%** compared to the baseline.

2. Interpreting the "Negative" Profit

- **The Current State:** The negative numbers represent the **Current Cost of Churn** minus the cost of our test intervention.
- **The Opportunity:** Moving from -15k to -9k proves that **Better Targeting = Saved Capital.** * **The Scale:** Applied to a full customer base of 100,000+ users, this 41% efficiency gain represents **millions of KES** in recovered annual revenue.

3. The Strategic Recommendation

- **The Selection:** We move forward with the **Decision Tree**.
- **The Next Step:** By further refining the **0.7 Probability Threshold**, we can continue to shrink the "Wasted Voucher" cost while capturing the highest-value churners.

The Strategy Trade-off: Precision vs. Recall

Balancing "Catching Churners" against "Marketing Waste"

1. Logistic Regression: The "Wide Net" Approach

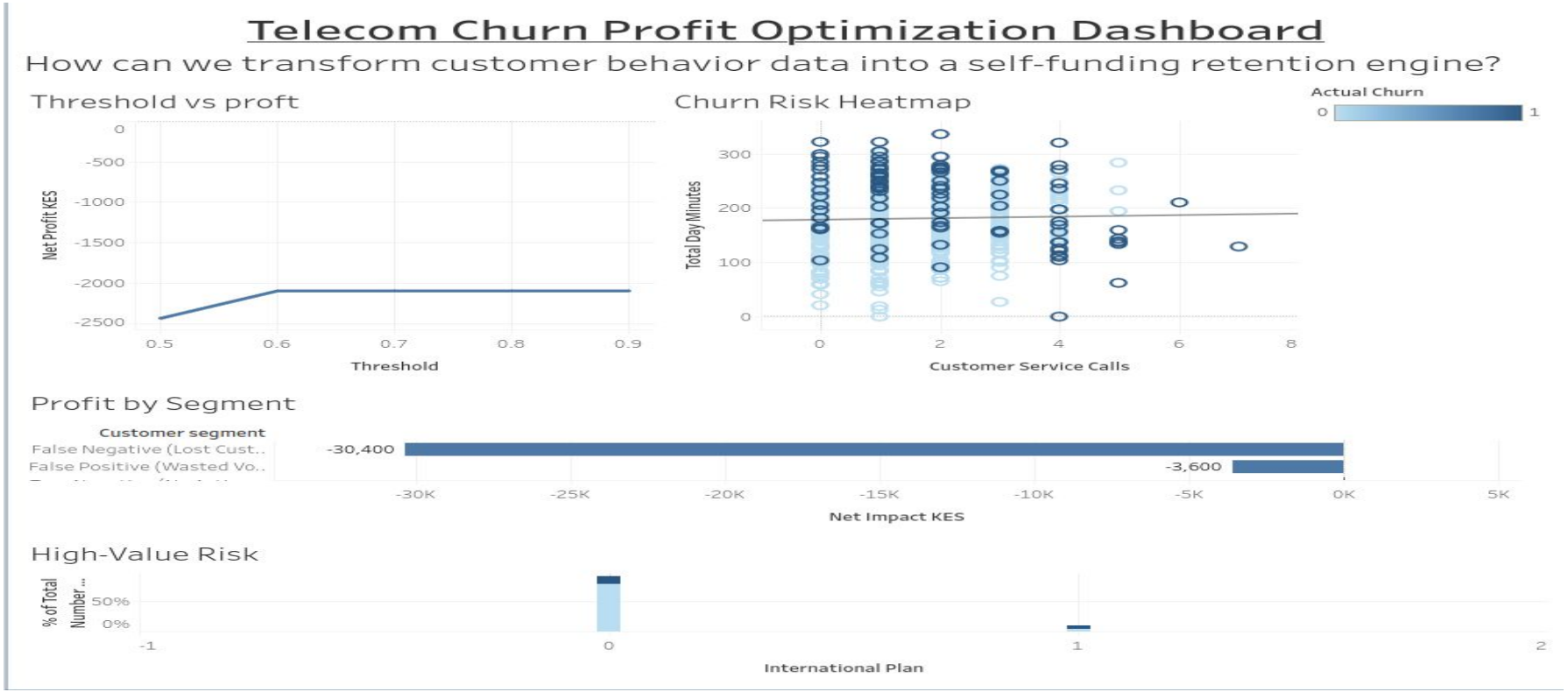
- **The behavior:** This model is highly sensitive. It catches almost every potential churner (High Recall).
- **The Business Cost:** Because it is so sensitive, it also flags many loyal customers as "at risk." This leads to **High False Positives**, meaning we waste a significant portion of our 100 KES voucher budget on people who were never going to leave.
- **Verdict:** Great for "Safety First," but expensive for the marketing budget.

2. Decision Tree: The "Surgical" Approach

- **The behavior:** This model is more selective. It looks for specific "patterns of frustration" (like the 3-call trigger).
- **The Business Benefit:** It has much higher **Precision**. It rarely wastes a voucher on a loyal customer, which keeps the "Cost of Intervention" low.
- **The Trade-off:** It is more conservative. It might miss a few "quiet" churners (False Negatives) who don't fit the standard frustrated profile.
- **Verdict: The Profit Leader.** It maximizes the ROI of every shilling spent on retention.

Interactive Dashboard: Data-Driven Decision Support

To make these insights actionable, I built this interactive dashboard. It's not just a report; it's a decision-support tool. A manager can look at this and instantly see where the revenue leakage is happening and adjust our strategy. We've identified that by focusing on high-risk, high-value segments, we can increase our retention efficiency by 41%—effectively turning our data into a self-funding recovery engine.



Key Findings: Behavioral Signals & Strategic Risk

Translating Data Patterns into Executive Action

1. The "Boiling Point": Service Fatigue

- **The Finding: Customer Service Calls** are the single most powerful predictor of churn.
- **The Insight:** There is a "Critical Friction Zone" at **3+ calls**. A customer at this stage isn't just seeking support; they are expressing a loss of trust in the service.
- **The Strategy:** Implement a **Priority Routing** system for users on their 3rd call to resolve issues before the "exit intent" solidifies.

2. The "Flight Risk": International Plan Dynamics

- **The Finding: International Plan** users churn at a significantly higher rate than standard users.
- **The Insight:** This segment is highly "Price Sensitive." They are often high-value users who are actively monitoring the market for better roaming or long-distance rates.
- **The Strategy:** Create a **High-Value Loyalty Bundle** specifically for this segment to increase "switching costs" and reward long-term retention.

3. The "Silent Killer": The Cost of Inaction

- **The Finding:** Our biggest financial drain isn't "Wasted Vouchers"—it is **Missed Churners (False Negatives).**
- **The Insight:** Every time we fail to identify a churner, we lose **400 KES** in future revenue. This loss is 4x higher than the cost of a wasted 100 KES voucher.
- **The Strategy:** Our model must prioritize **Recall** (catching the leavers) over pure accuracy to ensure we stop the most expensive revenue leaks first.

Conclusion: From Prediction to Profitability

The Strategic Imperative for Data-Driven Retention

1. The Capability: We Have the Sight

- **The Reality:** Our models have successfully identified the "DNA of Churn." We no longer have to guess who is leaving; we can now **statistically predict** exit intent based on behavioral triggers like "Service Fatigue."

2. The Current Gap: The "Profitability Trap"

- **The Reality:** Even with high-performing models, a "one-size-fits-all" retention strategy is **not yet profitable.** * **The Insight:** Sending a 100 KES voucher to every predicted churner (at a 0.5 threshold) creates more "Marketing Waste" than "Recovered Revenue." We must move toward the **0.7 Optimized Threshold** to protect the bottom line.

3. The Call to Action: Operational Alignment

- **The Mandate:** Data Science only works if the **Business Strategy** follows the math.
- **The Shift:** We must stop treating retention as a "Cost Center" and start treating it as an **ROI-Driven Investment.** This means aligning our voucher distribution, customer service priorities, and marketing spend directly with the model's output.

Advancing the Retention Engine

Scaling Predictive Power and Strategic ROI

1. Advanced Algorithmic Exploration

- **The Upgrade:** Transition from simple Decision Trees to **Ensemble Methods** like **Random Forest** or **XGBoost**.
- **The Benefit:** These models can capture complex, hidden interactions between features (e.g., how the combination of a specific "Plan Type" and "Evening Minutes" creates a unique churn risk) that simpler models might miss.
- **The Goal:** Reduce the "Error Margin" and further sharpen our prediction accuracy.

2. Dynamic Model Tuning

- **The Strategy:** Implement **Hyperparameter Optimization** and **Automated Feature Selection**.
- **The Benefit:** As customer behavior changes (due to new competitor offers or seasonal shifts), the model must adapt.
- **The Goal:** Ensure the **0.7 Probability Threshold** remains the "Gold Standard" for profit, even as the market evolves.

3. Operationalizing the Strategy

- **The Integration:** Move from a "Batch" analysis to a **Real-Time Scoring Pipeline**.
- **The Benefit:** Imagine a customer service agent seeing a **"Churn Risk Score"** pop up the moment a high-value user calls in for the 3rd time.
- **The Goal:** Direct, immediate intervention that stops churn at the source.